Remaining schematic work

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10 active issues. B62: red RX (TP5 above U3), black G (printed GND test hole). Report volts and whether steady or varying. No other supply or continuity readings need repeating. Q3 stays fitted. Completed steps are not new bench requests.

Drawing / connectivity	Remaining data	ERC
0 fitted open pads; 0 open PIC pads; 0 dangling wire ends. 82 native/model net partitions match.	44 ceramic values; LED1 footprint blank. Two local-only PIC branches remain explicit; unsupported R33 withdrawn.	1 retained U6/U6A output conflict. Five stock power-source flags retained; B59 RA0-TP6 join recorded.

Audit boundary

Native wire geometry, library definitions, all catalog pad assignments and existing evidence were checked. Zero open pads does not establish hidden copper or functional behavior. U6A is DNP; its regulator symbol and visible conflict remain by user choice. B59 supports the RA0-TP6 net merge. No new NC markers or ERC exclusions.

Order of work

Resolve hidden paths and device behavior: S1 contacts and the front capacitor-bank ANT2 connection are established. Q3 stays fitted with identification deferred. Two local-only PIC branches, candidate device roles and powered behavior remain. RA0-TP6 and TP14-OUTA are now resolved. B62 checks receiver supply at TP5 to interpret zero VREF. TP11/TP17 hidden continuation remains qualified; Q3 stays fitted.

Recover critical component values: 44 ceramic values; prioritize tuning and receiver after topology. Use the LCR meter with known test conditions and account for parallel components.

Finish interface and package mapping: LED colour branches and J5 board-contact/output mapping are established. LED footprint, connector fit and external motor harness orientation remain reproduction qualifications. Obtain a matching LED vendor drawing or pad geometry; qualify J5 and the other packages before laying out a replacement PCB.

Resolve hidden paths and device behavior

E03 - RF supply and Q8 drive

Established: C5/C6 parallel wiring and Q8 ground/output joins are established. B37 P-F = 1 ohm supports Q8.T2/native2 to C6.1 on H_RF_VDD; no fresh baseline or stability statement supplied. SIL3724A and the U2 inverter remain candidates. B54/B55 measured VSYS and RF supply at 6.3 V in the red-blinking bench state.

Remaining: Candidate gate/source roles, P-channel turn-off margin and RF timing remain unverified. The source-supply continuity question no longer requires a bench check.

Next: Later operating check on the original hardware: establish source/gate voltages and actual gate-to-source drive, then RF timing if suitable equipment is available. Do not repeat P-F continuity.

Close when: Source path and actual gate-to-source drive are supported; C6 value is handled by V01.

E04 - Antenna and tuning-cell topology

Established: PIC-to-control resistors and local tuning midpoints are mapped. B44/B45 midpoint-to-antenna DC readings are OL. B52 C11 outer end to ANT2 = 1.4 ohm; photo-shared outer pads of C14/C29/C44/C47 now join ANT2. Rear capacitor returns remain GND. ANT1-ANT2 is about 5 ohm with antenna fitted. The mistaken 414 kohm N-B report was corrected to B-G. Q3 L-S latest 0.48 V, L-R 2.3 V; all other diode directions OL.

Remaining: The drawn PNP B=R/E=L/C=S candidate is still not confirmed by in-circuit junction readings. Capacitor values and actual switching/resonance behavior remain unknown. The front capacitor-bank antenna connection is now established.

Next: No more antenna continuity sweep. Leave Q3 fitted as requested. Agent reviews candidate device roles against the corrected topology; isolate-device identification is deferred. RF operating checks follow only after values and suitable instrumentation are addressed.

Close when: Tuning device pin roles and switched RF behavior are supported; outer ANT2 path is already resolved.

E06 - PIC test branches: TP11 and TP17

Established: PIC14/RC3 reaches TP11; PIC18/RC7 reaches TP17. Front/rear photos show no onward surface trace at those test points. Both are valid endpoints, not dangling wires. RA0/C25 to TP6 measured 1.5 ohm in B59 and is now resolved on RX_A_FILTER.

Remaining: Only the possibility of additional inner-layer connections on RC3/TP11 and RC7/TP17 remains. A test-only purpose is plausible but not established by photographs.

Next: No justified new continuity pair is identified. Retain the two known PIC-to-test-point routes and the explicit inner-layer limitation; do not repeat TP16-to-TP11/TP17. Continue the prepared supply-voltage batch for the operating issues.

Close when: Additional continuation is traced, or physical layer evidence establishes these are test-only branches. Until then, reproduce the known local routes without asserting that hidden continuations are absent.

E07 - PIR: Q2 behavior and operating supply

Established: J3/U7 ordering, U6 external pads and PIC21/TP16/Q2 output routing are established. U6 pin5 is tied to VIN; pin4 is unused. B58 measured 3.3 V at J3.1 with PIR connected; logic VDD was 4.5 V.

Remaining: Q2 terminal functions, PIR raw-signal behavior and PIC input response remain unverified. The loaded PIR supply is established at 3.3 V in the reported state; transients and other states are not characterized.

Next: After the receiver supply check, compare J3.2 raw signal and TP16/PIC21 as PIR state changes. Do not repeat the 3.3 V baseline or established connector continuity.

Close when: Q2 signal behavior and U6 output are supported by device or operating evidence.

E08 - Receiver and Q7 operating behavior

Established: U3 input returns and R23/R42/C40, C23/R25/R48 and DNP R47/C42 paths are reconciled. VREF supplies U3 pin5. B59 C25.1-TP6.1 = 1.5 ohm establishes the receiver filter connection to PIC2/RA0; R27 remains between that node and U3.1. B57 measured the printed VREF test point at 0 V while the red LED blinked about once per second.

Remaining: Q7 transistor pin roles/supply switching, receiver gain/filter response and the functional MCU interface are unverified. H_RX_DETECT is only a series junction, not proof of a missing output wire. D3/R42-to-PIC25/RB4 wiring is established; its functional role belongs to receiver/Q1 operating validation, not an assumed missing wire. VREF at 0 V must be interpreted against PIC/receiver state; receiver supply and amplifier output voltages remain unmeasured.

Next: B62: red TP5 receiver-supply test point to black GND, DC volts in the same red-blinking state. This distinguishes absent receiver supply from a bias-path issue; no baseline rail repeats.

Close when: Receiver supply, bias and the existing RA0/RA1/RB4 interfaces have supported operating roles; no presumed extra output wire is required.

E09 - Supply voltages and Q1 control

Established: VSYS/RF 6.3 V, printed VDD 4.5 V, PIR supply 3.3 V with PIR attached; VREF 0 V and red LED approximately 1 Hz. TP14 is now established on U5.8/OUTA at 1.5 ohm; R43.2/V015 follows its existing TP14 association. BATTERY+ stays separate. TP14-Q1 gate measured 516 ohm.

Remaining: Q1 control behavior, motor-output states, the path giving 516 ohm from OUTA to Q1 gate, zero VREF and the red-flashing state remain unresolved. R3/R4/C28-to-RB3 functional role is inferred.

Next: First B62: TP5 receiver supply to GND in the same powered state. Then choose operating-state checks for Q1 and the bridge, using the corrected TP14/OUTA assignment; no repeat continuity requested.

Close when: Actual supply levels and control behavior are recorded without merging battery, VSYS or filtered branches.

E10 - D6 breakdown and ICSP compatibility

Established: D6 anode is GND and cathode is VPP; forward drop is about 0.7V and reverse diode mode is OL. J1 pin order is resolved. Primary programming specification DS40001753B requires 8.0-9.0 V for high-voltage entry; low-voltage entry exists only when LVP is enabled.

Remaining: G3 is nonunique: MMSZ5228BS is 3.9 V; PZU2.4B is 2.4 V. Neither rating has been measured on this board. A directly connected clamp of either rating would load the required high-voltage ICSP level. The current PIC LVP configuration and actual D6 breakdown are unknown.

Next: No repeated continuity or diode-mode check. Marking research establishes ambiguity, not a rating. Keep D6 fitted for now; resolve its rating by an isolated current-limited test only if needed. Before any programmer session, establish the supported entry method; do not apply a breakdown test through the PIC.

Close when: Clamp class/rating is established or explicitly bounded for reconstruction, and programming compatibility is checked.

I01 - Candidate device identities and pin roles

Established: Manufacturer candidate datasheets and pin tables exist. D2 local wiring is complete; measured R -> S -> L diode behavior supports a series pair.

Remaining: Exact fitted identity/ratings remain candidates for D1-D6, Q1-Q8/Q11, U1/U2/U6. The 4P marking alone does not identify three-pin D2; a manufacturer 4P code for a two-pin diode is incompatible.

Next: Resolve electrical risks through E03/E04/E07/E08/E09/E10. For remaining ambiguity retain the supported equivalent circuit and disclose ratings not established; no sharper-marking request or repeat D2 rail sweep. Exact manufacturer recovery is not always possible with a meter.

Close when: Each part has a supported pin/function model and adequate ratings or an explicit accepted-equivalent limitation; exact branding is not falsely claimed.

Recover critical component values

V01 - Capacitor values and passive ratings

Established: C5 is 470uF/16V. L1/L2 readings are 1.4/2.2uH. Retained marked resistor nominal values are recorded.

Remaining: 44 fitted ceramic values remain unknown. L1/L2 measurement conditions and component ratings are not established.

Next: After tuning topology, prioritize antenna/RF/receiver capacitors with the LCR meter. Record frequency, mode and isolation. Parallel parts give a combined reading; lift one terminal only where an individual value is needed. Report existing L1/L2 measurement conditions without remeasuring yet.

Close when: Critical individual values are recovered; any bypass substitutions and unverified ratings are explicitly identified assumptions.

All unrecovered ceramic values

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

Finish interface and package mapping

M01 - LED footprint and mechanical qualification

Established: LED1 is the only unassigned footprint; R30/PIC23 lights red and R31/PIC22 lights green. C5 diameter/height, S1 body/actuator size, R8 and C11 sizes are already measured. J5 rear placement and both board contacts are mapped to U5 outputs from front/rear photos.

Remaining: LED1 land pattern/manufacturer numbering and internal pairing of its externally grounded cathode pads remain unverified. C5 pitch and S1 pad fit/3D heights are reproduction qualifications, not unknown schematic wires. J5 stock-footprint pad order/pitch and external motor harness colours remain unverified.

Next: Obtain LED underside/pad geometry or a matching vendor drawing before selecting a footprint. For later PCB reproduction check C5 lead pitch and S1 pad fit; use actual 16mm capacitor and 6mm switch heights. Do not repeat body dimensions. Match J5 physical A/B contacts to the chosen connector drawing before PCB layout; record harness colours when available, without repeating visible board continuity.

Close when: LED footprint has a supported pad map; remaining package-fit assumptions are documented separately from schematic connectivity.

PCB reproduction requirements

Electrical design

Close the 10 active main-board issue groups, recover critical values, and qualify candidate pin functions/ratings.

Agent: Trace visible copper, review datasheets and keep the model/native files consistent.

Owner input: Antenna outer-rail continuity complete. Future measurements are selected individually after photo review; powered behavior remains unverified.

Mechanical fit

Dimensioned board perimeter, mounting holes, connector positions and lead geometry; LED land pattern.

Agent: Create a dimensioned replacement layout from supported measurements.

Owner input: Only geometry not recoverable reliably from the photos; existing body dimensions are retained.

Copper and stackup

No routed PCB exists; four layers remain a hypothesis. Continuity identifies nets, not hidden track geometry or dielectric thickness.

Agent: Design and check a replacement routing/stackup once electrical and mechanical data are sufficient.

Owner input: Layer/board thickness evidence if exact physical replication is required.

Firmware and configuration

No PIC firmware/configuration backup is present. An identical populated board needs the program and configuration, or compatible replacement firmware.

Agent: Plan a read-only backup after ICSP/D6 compatibility is established; do not erase the original.

Owner input: Programmer availability later. Protection may prevent reading the program; this is currently unknown.

Complete working assembly

Powered validation, motor/antenna/PIR integration, and daughterboard internals if that board is also being copied.

Agent: Define staged checks against the original and validate the replacement.

Owner input: Operating measurements on the original; external assemblies and their geometry if reproduced.

Scope and sources

Main-board schematic finalization. Routed PCB/layer stack, PIR daughterboard internals and firmware reconstruction remain later work, not completed. Empty-option population design is not needed to reproduce the observed DNP board.

DNP options retain all modeled local routes and candidate roles. J6.2/R36.1 terminates at an unpopulated external interface; no additional onboard rail is assumed. U6A stays a regulator symbol with its visible output conflict by explicit user choice.

All previous issue dispositions: evidence/finalization_review_v0931.json. Native geometry and uncertainty coverage: evidence/finalization_audit.json. Current register: evidence/remaining_work.json. Archived observations and old checklists: docs/history/README.md.

Power-source flags describe supply paths through existing passive components; they do not prove voltage or transistor operation. Installed KiCad 10 libraries and project symbols match the embedded pin/graphic definitions. GUI library reload has not been verified.

Reference rails

Rail	Existing modeled path
BATTERY+	Actual positive battery terminal -> Q1 single pad/native3 (candidate drain). TP14 is OUTA, separate from this rail.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.
VREF	TP104, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.
GND	Battery negative and ground copper.